

AFRA AZAD

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EDUCATION

Bachelor of Arts and Science, University of Toronto St. George — Data Science
Dean Scholar List — CGPA: 3.91

Sep 2021 to Dec 2026
Later Life Learning OSOTF Award

SKILLS

Machine Learning: Machine Learning, Deep Learning, NLP, Transformer Models (BERT), Transfer Learning
Programming: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch), R, SQL/MySQL
Data & Analytics: Power BI (DAX), Apache Spark, MS Excel, MS Access
Cloud & Tools: MS Azure, MS Power Automate, Git, Unix, Anaconda, APIs
Web: HTML, CSS, JavaScript, Web-scraping

EXPERIENCE

Data Analyst Intern
Culligan Quench

June 2024 – August 2024, June 2025 – August 2025
Vaughan, ON

- Performed large-scale **data quality assessment**, data validation, and preprocessing on 8,000+ structured records using **Python, Pandas** from **MS Azure** database.
- Developed an end-to-end data pipeline from a legacy data warehouse to **Power BI** using **Python API**, improving data accessibility for analytics and business reporting.
- Implemented **SQL** queries to support early-stage **database development** in **MS Azure** database to improve data accessibility for analytics teams.
- Applied fuzzy matching and **geospatial analysis** using the **Google Maps API** to join the technician and sales datasets based on geocoded addresses.
- Developed automated **Power BI** dashboards integrating financial, sales, inventory, and customer data to support reporting and analysis for business stakeholders.
- Conducted **statistical analysis** through hypothesis testing to evaluate whether revenue differed significantly across company sectors (agricultural, financial, and technology), finding no statistically significant differences.
- Clustered 10,000+ customer locations using unsupervised centroid-based clustering for optimal delivery zones in **R**.
- Conducted external **data merging, cleaning and wrangling** of government APIs and commercial datasets (e.g. Dun & Bradstreet) with around hundred thousand companies to support **market analysis**.
- Presented dashboards, analytical findings, and business insights to managers during weekly meetings.

Student Data Analyst

University of Toronto Temerty Faculty of Medicine

October 2024 – April 2025
Toronto, ON

- Evaluated existing **NLP Named Entity Recognition (NER)** models using SpaCy, LLaMA, and Generative AI to de-identify **Personally Identifiable Information (PII)** over 33 thousand for data privacy and security in **Python**.
- Performed content analysis on **embedded** survey comments using **transformer-based architectures BERT** for feedback over time, utilizing **GitHub, Unix, and Anaconda** for version control, scripting, and remote computing.
- Applied **transfer learning** by fine-tuning **BERT transformer models** for **sentiment analysis**.
- Implemented raw text data pipelines including **tokenization, feature extraction, and data cleaning** for unstructured qualitative comment data.
- Optimized classification models using **GridSearchCV** and **TF-IDF vectorization**, achieving **98.5% accuracy**.

Machine Learning Developer
UHN (University Health Network)

September 2025 – Present
Toronto, ON

- Performed large-scale **data-cleaning** of **raw data** through unsupervised techniques such as outlier detection, missing data handling, distribution analysis, and **PCA-based** dimensionality reduction on 10,000+ patient records.
- Designed and evaluated machine learning models for longitudinal predictions, testing various approaches.
- Tested **machine learning** modes, such as, LSTM, BiLSTM, GRU, RNN, and 1D-CNN architectures with research assistance using **Claude AI**.
- Evaluated models such as **timeseries models**, **XGBoost**, GMM, Lasso, and Ridge, combined with **stacking** and **blending**.
- Implemented models using **Python**, leveraging **NumPy**, **Pandas**, Scikit-Learn, and TensorFlow/Keras.
- Conducted extensive model optimization, hyperparameter tuning and evaluated over 5,000 model configurations using multiple performance metrics including: **RMSE**, **MAE**, **MSE**, **R²**.
- Developed a production-ready predictive modelling pipeline and automated updates to **SQL databases**.
- Conducted **experimental design** through answering what model best forecasts anxiety scores with **80-20 training split**.
- Documented current Python scripts to the latest Python version 3.14 through **Git pull requests**, code reviews and branching.
- Supported the clinical research team by translating predictive modelling results into a **final report** covering 10,000+ patient records.

OpenAi Project Lead

UTMIST (University of Toronto Machine Intelligence Student Team)

Sep 2024 - April 2025
Toronto, ON

- Implemented an image captioning model for NVDA (NonVisual Desktop Access) for those who are visually impaired.
- Conduct **literature search** for pre-existing image caption machine learning models, and developed new models by extracting over 500 thousand images and captions in **Python**, and parsed the captions datasets from **MS Access**.
- Developed website with **HTML**, **CSS**, and **JavaScript** to demonstrate image captioning function.
- Thoroughly trained, validated and tuned **LLM** and **CNN** models, using libraries (**numpy**, **Keras**, **tensorflow**) and utilizing **Git**, **Python**, **Pandas** and **Numpy** in local **virtual environment** in Google Collab.
- Managed a team of 6 members, coordinating tasks, setting goals, and facilitating clear communication to ensure the project was completed by the deadline.
- Organized and **led weekly team meetings** to share project updates, align priorities, and coordinate work across team members.
- Communicated technical concepts and project updates to non-technical stakeholders.
- Delegated responsibilities and adapted to challenges in model publishing, maintaining team organization and accountability in a fast-paced environment.

Products and Technology Summer Intern

PwC (PricewaterhouseCoopers)

May 2023 - August 2023
Toronto, ON

- Individually developed four automated **Power BI** data visualization reports on quantitative data such as team utilization and financial costs using relational databases.
- Applied data analysis and presented reports to various **stakeholders**, including managers, associates, and directors.
- Collaborated with managers and guest speakers in organizing and presenting bi-weekly team meetings for 50.
- **Managed six projects** in a fast-paced work environment, multitasking, adapting to different environments, project requirements, and meeting tight deadlines and business requirements.
- Manipulated **MS Excel** functions to conceptualize, develop, and optimize a predictive linear regression **statistical model** to estimate the budget required for **UX design** projects to solve inaccurate budgeting proposals.